



جامعة ٦ أكتوبر التكنولوجية
6 OCTOBER TECHNOLOGICAL UNIVERSITY



6 October Technological University

Faculty of Industry and Energy Technology

Information Technology Program

Smart IoT Attendance System Ecosystem

An Integrated RFID-Based Academic Management Solution

Graduation Project

Submitted in Partial Fulfillment of the Requirements
for the Degree of Professional Bachelor's Degree in
Information Technology

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Abstract

The Smart IoT Attendance System is a comprehensive, offline-capable academic management solution designed to modernize and streamline the attendance tracking process within university environments. Developed as a graduation project at the University of 6 October Technology, College of Computer Science and Information Technology, this system addresses the limitations of traditional attendance methods by leveraging Internet of Things (IoT) technologies, specifically Radio Frequency Identification (RFID), combined with a robust software ecosystem.

The system architecture comprises four integrated components: an ESP32-based IoT device equipped with RC522 RFID readers for physical card scanning, a FastAPI-powered backend server handling business logic and data persistence, a React-based web dashboard providing administrative interfaces, and a Kotlin Android application offering mobile accessibility for students and instructors. The entire system is deployed on a Raspberry Pi 4, creating a self-contained, offline-first infrastructure that operates independently of internet connectivity.

Key features include multi-faculty support with role-based access control (RBAC) spanning six hierarchical levels, real-time attendance recording with duplicate prevention, comprehensive gradebook management with a 40/60 attendance-to-assessment weighting system, AI-powered academic insights using Retrieval-Augmented Generation (RAG) techniques, and automated reporting capabilities. The system supports three faculties — Engineering, Medicine, and Business — with over 100 active students and multi-department seeding.

The project demonstrates the successful integration of hardware and software components to create a reliable, scalable, and user-friendly attendance management system. Testing results confirm the system's ability to handle 100+ concurrent users, maintain data integrity through unique attendance constraints, and provide sub-second response times for RFID-based attendance recording. The offline-first design ensures continuous operation in environments with limited or no internet access, making it particularly suitable for deployment in academic institutions across Iraq.

Chapter 1: Introduction

1.1 Background

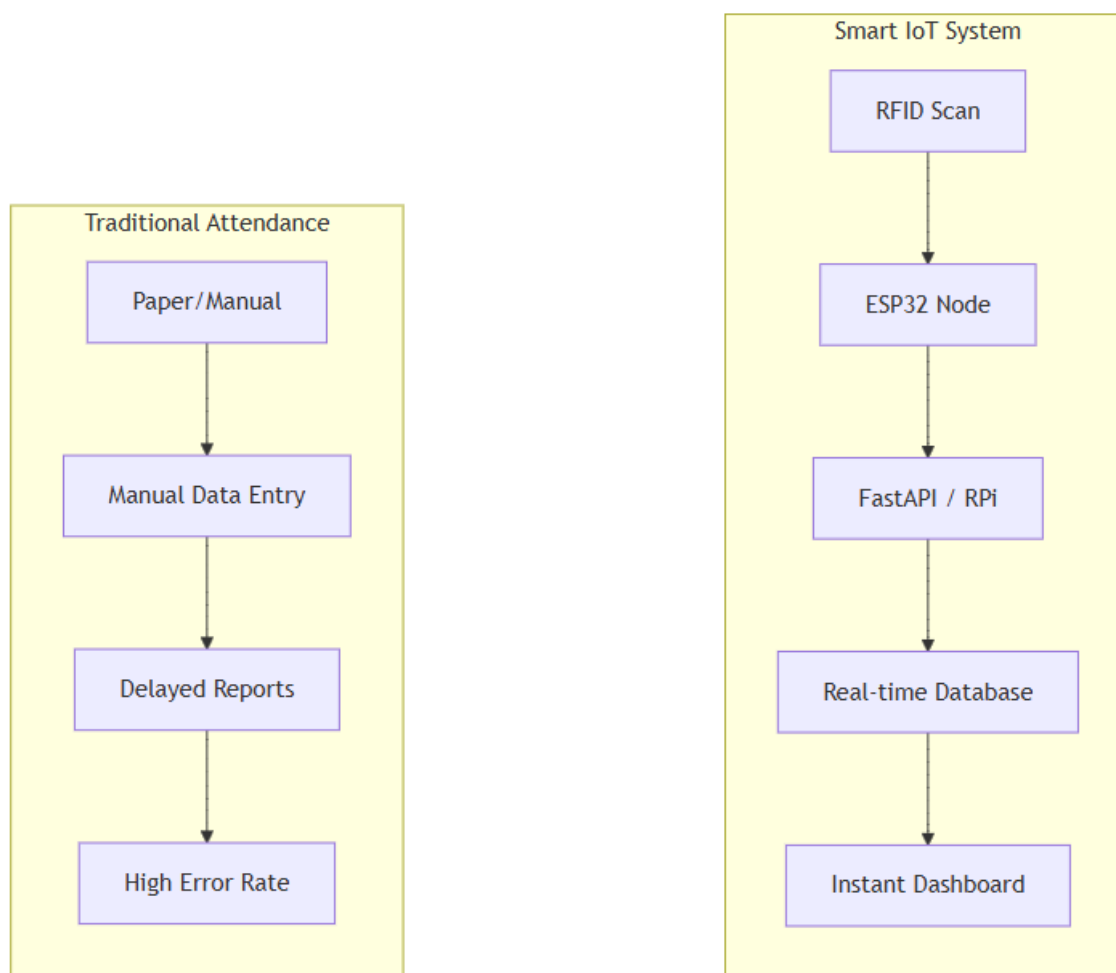
The management of student attendance has long been a critical yet challenging aspect of academic administration in higher education institutions. Traditional methods of attendance tracking, including manual roll calls, paper-based sign-in sheets, and simple electronic spreadsheets, have proven to be time-consuming, error-prone, and susceptible to fraudulent practices such as proxy attendance. These conventional approaches not only consume valuable instructional time but also fail to provide real-time insights into student engagement and performance patterns.

The advent of the Internet of Things (IoT) has opened new avenues for automating and optimizing attendance management processes. IoT refers to the network of physical devices embedded with sensors, software, and connectivity capabilities that enable these objects to collect and exchange data. In the context of educational institutions, IoT technologies — particularly Radio Frequency Identification (RFID) — offer a compelling solution for creating automated, accurate, and efficient attendance tracking systems.

Radio Frequency Identification (RFID) technology uses electromagnetic fields to automatically identify and track tags attached to objects. In an attendance system context, students are issued RFID cards or tags that can be scanned by readers installed in classrooms or lecture halls. This approach eliminates the need for manual intervention, reduces errors, and provides instant verification of student presence.

The University of 6 October Technology, like many academic institutions in Iraq, faces unique challenges in implementing technology solutions. Limited internet infrastructure, intermittent connectivity, and the need for cost-effective deployment models necessitate an offline-first approach to system design. This project addresses these constraints by developing a self-contained attendance management system that operates entirely on local infrastructure without dependence on external internet connectivity.

Figure 1.1: Traditional attendance methods vs. Smart IoT Attendance System



1.2 Problem Statement

Despite the availability of various attendance management solutions in the market, several significant challenges persist in the context of Iraqi higher education institutions:

- 1. Internet Dependency:** Most modern attendance systems require continuous internet connectivity for cloud synchronization, making them unsuitable for deployment in environments with limited or unreliable internet access.
- 2. Manual Process Inefficiency:** Traditional manual attendance methods consume 5-10 minutes of each lecture session, accumulating to significant instructional time loss over an academic term.

3. **Proxy Attendance:** Paper-based and manual electronic systems are vulnerable to proxy attendance practices, where students sign in on behalf of absent peers, compromising data integrity.
4. **Data Fragmentation:** Attendance records maintained separately by individual instructors create fragmented datasets that are difficult to aggregate for institutional analysis and reporting.
5. **Limited Analytics:** Conventional systems provide raw attendance counts without analytical capabilities to identify at-risk students, attendance patterns, or performance correlations.
6. **Multi-Faculty Complexity:** Large universities with multiple faculties, departments, and academic programs require attendance systems capable of handling complex organizational hierarchies and varying academic structures.

1.3 Project Objectives

The primary objective of this project is to design and implement a comprehensive Smart IoT Attendance System that addresses the identified challenges through the following specific goals:

- Develop an offline-capable attendance management system that operates without internet connectivity, suitable for deployment in Iraqi academic institutions.
- Implement RFID-based student identification using ESP32 microcontrollers and RC522 readers for automated, accurate attendance recording.
- Create a robust backend server using FastAPI with role-based access control supporting six hierarchical user roles from Super Admin to Student.
- Build a responsive web dashboard using React for administrative functions, data visualization, and system management.
- Develop a native Android application using Kotlin and Jetpack Compose for mobile access to attendance records and academic information.
- Implement a comprehensive gradebook system with configurable attendance-to-assessment weighting (40/60 distribution).

- Integrate AI-powered analytics using RAG techniques to provide academic insights and early warning systems for at-risk students.
- Design a portable deployment architecture using Raspberry Pi 4 that enables rapid system setup at new university locations.

1.4 Project Scope

The scope of this project encompasses the design, development, and deployment of a complete attendance management ecosystem. The system boundary includes:

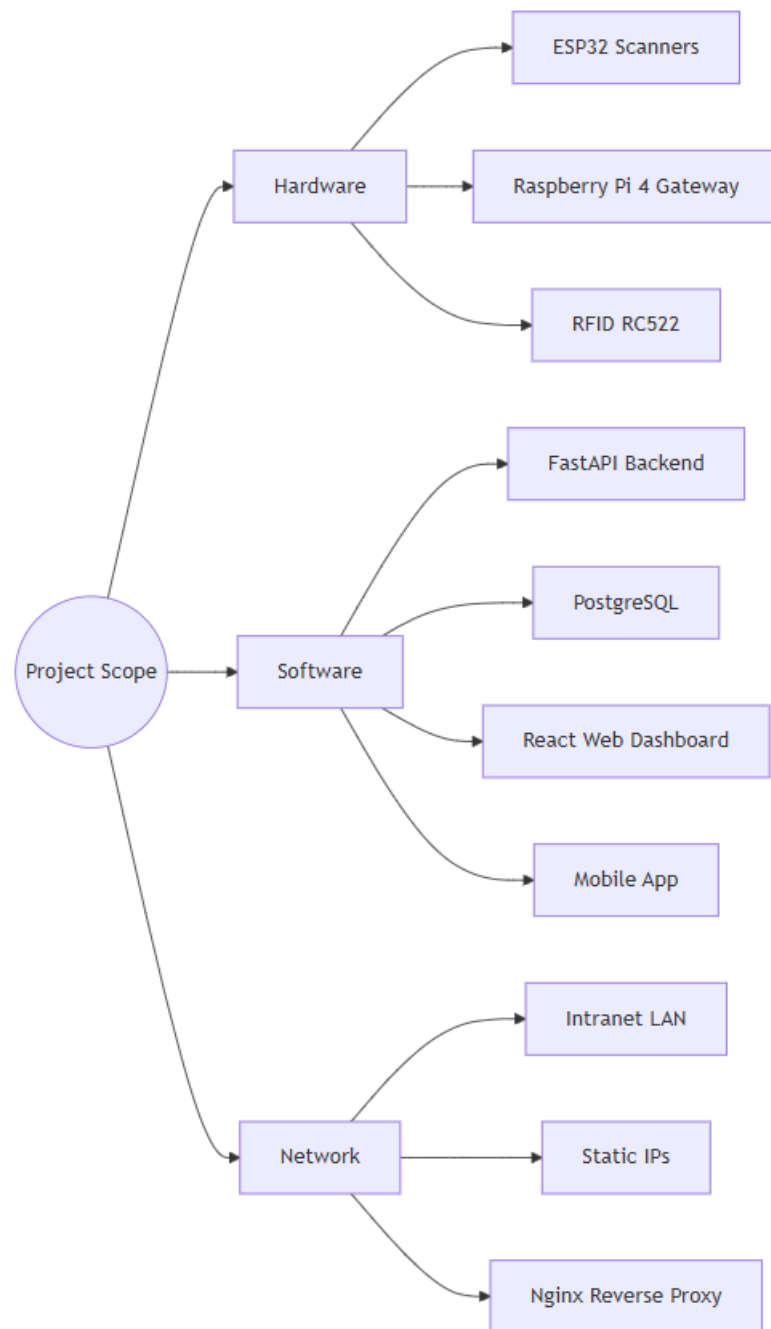
1.4.1 In Scope

- RFID-based attendance scanning and recording for lecture sessions, labs, and examinations.
- Multi-faculty, multi-department, and multi-course academic structure management.
- User management with six-tier role-based access control (Super Admin, Admin, Doctor, Instructor, Engineer, Student).
- Web-based administrative dashboard with reporting, analytics, and system configuration.
- Android mobile application for students and instructors.
- Gradebook management with weighted assessment and attendance scoring.
- AI-powered academic query system for institutional insights.
- Offline-first deployment on Raspberry Pi 4 with local Wi-Fi access point.

1.4.2 Out of Scope

- Biometric authentication (fingerprint, facial recognition) — the system uses RFID cards exclusively.
- iOS mobile application development.
- Cloud-based deployment or multi-campus synchronization.
- Integration with existing university ERP systems.
- Payment or financial management features.

Figure 1.2: Project scope diagram



1.5 Significance of the Project

This project holds significant value at multiple levels:

1.5.1 Academic Significance

The project demonstrates the practical application of IoT concepts, embedded systems programming, full-stack web development, mobile application development, and artificial

intelligence integration within a unified system. It serves as a comprehensive example of how modern software engineering practices can be applied to solve real-world problems in educational contexts.

1.5.2 Practical Significance

For the University of 6 October Technology and similar institutions, the system provides an immediately deployable solution that eliminates manual attendance overhead, improves data accuracy, enables data-driven academic interventions, and creates a foundation for digital transformation of academic administrative processes.

1.5.3 Technical Significance

The offline-first architecture, containerized deployment on Raspberry Pi, and integration of diverse technology stacks (Python/FastAPI, React, Kotlin, ESP32/Arduino) demonstrate best practices for building resilient, portable IoT systems in resource-constrained environments.

Chapter 2: Literature Review

2.1 Overview of Attendance Systems

Attendance tracking systems have evolved significantly over the past decades, transitioning from purely manual methods to sophisticated automated solutions. Understanding this evolution provides context for the design decisions made in this project.

2.1.1 Manual Attendance Systems

The earliest and still most common method of attendance tracking involves instructors calling out student names or passing around sign-in sheets. While simple to implement, these methods suffer from several well-documented deficiencies. Studies have shown that manual roll calls consume an average of 5-10 minutes per lecture session, which accumulates to approximately 8-16 hours of lost instructional time per course per semester. Additionally, manual systems are susceptible to human error in record-keeping and vulnerable to proxy attendance fraud.

2.1.2 Barcode-Based Systems

Barcode-based attendance systems represent an early attempt at automation. Students are issued barcode cards that are scanned at entry points using laser barcode readers. While more efficient than manual methods, barcode systems require line-of-sight scanning, are sensitive to physical card damage, and offer limited storage capacity on the card itself. These limitations have led to their gradual replacement by more robust technologies.

2.1.3 Biometric Systems

Biometric attendance systems utilize unique physiological characteristics for identification, including fingerprints, facial recognition, iris patterns, and voice recognition. These systems offer high security and eliminate the possibility of proxy attendance. However, they require expensive specialized hardware, raise privacy concerns, and can be affected by environmental factors such as lighting conditions or skin conditions. The cost and complexity of biometric systems make them impractical for large-scale deployment in resource-constrained educational environments.

2.1.4 RFID-Based Systems

RFID-based attendance systems have emerged as a balanced solution that offers automation, reliability, and cost-effectiveness. RFID technology enables contactless identification through radio waves, allowing students to simply tap their cards near a reader for instant verification. The technology is mature, with inexpensive tags and readers readily available, making it suitable for large-scale deployment in educational settings.

2.2 RFID Technology

Radio Frequency Identification (RFID) is a wireless technology that uses electromagnetic fields to automatically identify and track tags attached to objects. An RFID system consists of three primary components: tags (transponders), readers (interrogators), and an antenna system.

2.2.1 RFID Frequency Bands

RFID systems operate across several frequency bands, each with distinct characteristics:

- Low Frequency (LF, 125-134 kHz): Short read range (up to 10 cm), not sensitive to interference, commonly used for access control and animal identification.
- High Frequency (HF, 13.56 MHz): Moderate read range (up to 1 meter), includes NFC (Near Field Communication), widely used in contactless payment and identification cards. The RC522 module used in this project operates at this frequency.
- Ultra-High Frequency (UHF, 860-960 MHz): Long read range (up to 12 meters), used in supply chain management and inventory tracking.

2.2.2 RFID Tag Types

RFID tags are classified into three categories based on their power source:

- Passive Tags: Powered by the electromagnetic field from the reader, no internal battery, inexpensive, limited read range, suitable for short-range applications like attendance tracking.
- Active Tags: Contain their own power source (battery), longer read range, more expensive, used for asset tracking over large areas.

- Semi-Passive (Battery-Assisted) Tags: Use battery for chip operation but rely on reader for communication, offering a middle ground between passive and active tags.

Figure 2.1: RFID system components diagram



2.3 IoT in Education

The Internet of Things (IoT) has found numerous applications in the education sector, transforming traditional learning environments into smart, connected spaces. IoT in education encompasses a broad range of applications including smart classrooms, automated attendance, environmental monitoring, asset tracking, and student safety systems.

2.3.1 Smart Classroom Applications

IoT-enabled smart classrooms integrate various sensors and devices to create adaptive learning environments. Temperature and humidity sensors maintain optimal comfort levels, occupancy sensors manage lighting and HVAC systems efficiently, and interactive displays enhance student engagement. These systems collectively contribute to improved learning outcomes while reducing operational costs.

2.3.2 Student Monitoring and Safety

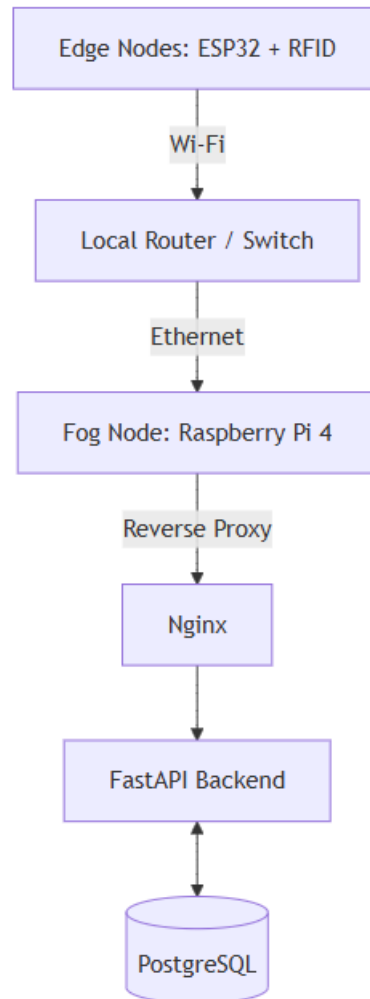
IoT-based student monitoring systems track student location and movement patterns within campus boundaries, enhancing safety and security. Wearable devices equipped with RFID or GPS capabilities enable real-time location tracking, while automated attendance systems provide accurate records of student presence in scheduled activities.

2.3.3 Data-Driven Decision Making

The data generated by IoT systems in educational settings provides valuable insights for institutional decision-making. Attendance patterns correlated with academic performance can identify at-risk students early, enabling timely intervention. Resource utilization data helps

optimize classroom allocation and scheduling, while long-term trend analysis supports strategic planning.

Figure 2.2: IoT architecture in educational environments



2.4 Related Work

Several research projects and commercial products have addressed the challenge of automated attendance management. This section reviews notable examples and identifies gaps that this project aims to fill.

2.4.1 RFID-Based University Attendance Systems

Numerous studies have explored RFID-based attendance systems for educational institutions. A study by Sharma et al. (2018) demonstrated an Arduino-based RFID attendance

system with SMS notification capabilities. While functional, the system lacked integration with academic management features and required internet connectivity for notifications. Similarly, research by Khan et al. (2019) presented a Raspberry Pi-based system with web interface, but did not address multi-faculty organizational structures or role-based access control.

2.4.2 IoT-Integrated Academic Management

Recent research has explored deeper integration between IoT devices and academic management systems. The work by Patel and Gupta (2020) incorporated temperature monitoring and automatic fan control alongside attendance tracking, demonstrating the potential for multi-function IoT deployments. However, their system relied on cloud-based infrastructure, limiting applicability in offline environments.

2.4.3 Mobile-First Attendance Solutions

The proliferation of smartphones has motivated research into mobile-based attendance solutions. Studies by Al-Tae et al. (2021) explored QR-code based attendance using mobile applications, while others investigated Bluetooth beacon technology for proximity-based attendance verification. These approaches reduce hardware costs but introduce new challenges related to device availability, battery consumption, and potential for gaming the system.

2.5 Comparison of Existing Systems

The following table presents a comparative analysis of existing attendance management systems and their features:

System	Offline	RFID	Mobile	RBAC	AI
Sharma et al.	No	Yes	No	No	No
Khan et al.	Partial	Yes	No	Basic	No
Patel & Gupta	No	Yes	No	No	No
Al-Tae et al.	Yes	No	Yes	No	No
This Project	Yes	Yes	Yes	6-Tier	Yes

Table 2.1: Comparison of existing attendance systems

The analysis reveals a clear gap in the market: no existing solution combines offline-first architecture, multi-faculty RBAC, comprehensive gradebook integration, AI-powered

analytics, and mobile application support in a single deployable package. This project addresses this gap by integrating all these capabilities into a unified system designed specifically for deployment in environments with limited internet infrastructure.

Chapter 3: System Analysis & Requirements

3.1 Functional Requirements

Functional requirements define the specific behaviors and functions that the system must perform. Based on stakeholder analysis and domain research, the following functional requirements have been identified:

3.1.1 User Management

- FR-UM-01: The system shall support user registration with role assignment (Super Admin, Admin, Doctor, Instructor, Engineer, Student).
- FR-UM-02: The system shall enforce password complexity requirements and secure storage using bcrypt hashing.
- FR-UM-03: The system shall implement JWT-based authentication with token expiration and refresh capabilities.
- FR-UM-04: The system shall support user profile management including personal information and role-specific settings.
- FR-UM-05: The system shall implement account blacklisting to prevent access for suspended users.

3.1.2 Academic Structure Management

- FR-ASM-01: The system shall support multi-faculty organization (Engineering, Medicine, Business, and expandable).
- FR-ASM-02: The system shall manage department hierarchies within each faculty.
- FR-ASM-03: The system shall manage course definitions including credit hours, prerequisites, and assigned departments.
- FR-ASM-04: The system shall support academic session creation with scheduling (date, time, location).
- FR-ASM-05: The system shall manage student enrollment in courses with section assignments.

3.1.3 Attendance Management

- FR-AM-01: The system shall record attendance via RFID card scanning using ESP32 devices.
- FR-AM-02: The system shall enforce one-scan-per-student-per-session uniqueness (409 Conflict on duplicates).
- FR-AM-03: The system shall validate RFID scans against active sessions and enrolled students.
- FR-AM-04: The system shall provide real-time attendance monitoring during active sessions.
- FR-AM-05: The system shall generate attendance reports by student, course, session, date range, and faculty.
- FR-AM-06: The system shall calculate attendance percentages and integrate with gradebook calculations.

3.1.4 Assessment & Gradebook

- FR-AG-01: The system shall implement a 40/60 attendance-to-assessment weighting distribution.
- FR-AG-02: The system shall support multiple assessment types (quizzes, assignments, midterms, finals) with configurable weights.
- FR-AG-03: The system shall validate that assessment weights sum to 100% within each course.
- FR-AG-04: The system shall provide gradebook finalization workflow with commit and rollback capabilities.
- FR-AG-05: The system shall calculate final grades automatically based on attendance and assessment scores.

3.1.5 IoT Device Management

- FR-IOT-01: The system shall support ESP32 device registration and mapping to specific classrooms/locations.

- FR-IOT-02: The system shall implement automatic RFID discovery mode for new card registration.
- FR-IOT-03: The system shall monitor device connectivity and report offline devices.
- FR-IOT-04: The system shall provide device configuration management including Wi-Fi settings and API endpoints.

3.1.6 Reporting & Analytics

- FR-RA-01: The system shall provide AI-powered academic insights using RAG-style query processing.
- FR-RA-02: The system shall generate at-risk student reports based on attendance and performance patterns.
- FR-RA-03: The system shall support data export to Google Sheets for external analysis.
- FR-RA-04: The system shall provide system health monitoring and telemetry dashboards.

Category	Key Requirements
User Management	Registration, JWT auth, RBAC, blacklisting
Academic Structure	Faculties, departments, courses, sessions
Attendance	RFID scanning, uniqueness, real-time monitoring
Gradebook	40/60 weighting, assessments, finalization
IoT	Device registration, discovery, monitoring
Analytics	AI insights, at-risk reports, data export

Table 3.1: Functional requirements summary

3.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and constraints of the system:

3.2.1 Performance Requirements

- NFR-P-01: RFID attendance recording shall complete within 2 seconds of card presentation.
- NFR-P-02: The system shall support 100+ concurrent users on the local Wi-Fi network.

- NFR-P-03: Web dashboard pages shall load within 3 seconds on standard hardware.
- NFR-P-04: API response times shall not exceed 500ms for 95% of requests under normal load.

3.2.2 Reliability Requirements

- NFR-R-01: The system shall auto-start on Raspberry Pi boot via PM2 process manager.
- NFR-R-02: The system shall maintain data integrity through database constraints and transaction management.
- NFR-R-03: The system shall implement automatic database migrations via Alembic.

3.2.3 Security Requirements

- NFR-S-01: All API endpoints shall require authentication except explicitly public routes.
- NFR-S-02: Role-based access control shall enforce data scoping at the API level.
- NFR-S-03: Passwords shall be hashed using bcrypt with appropriate work factors.
- NFR-S-04: JWT tokens shall have configurable expiration times and secure signing.

3.2.4 Usability Requirements

- NFR-U-01: The web dashboard shall be fully responsive across desktop, tablet, and mobile browsers.
- NFR-U-02: The Android app shall follow Material Design 3 guidelines.
- NFR-U-03: RFID scanning shall require minimal training (tap-and-go operation).

3.2.5 Deployment Requirements

- NFR-D-01: The system shall operate without internet connectivity (offline-first).
- NFR-D-02: The entire stack shall deploy on a single Raspberry Pi 4.
- NFR-D-03: Deployment shall use containerized processes managed by PM2.
- NFR-D-04: New university deployments shall be configurable through a startup wizard.

Category	Requirements
Performance	<2s RFID, 100+ users, <3s page load
Reliability	Auto-start, data integrity, migrations
Security	JWT auth, RBAC, bcrypt, token expiry
Usability	Responsive, Material Design, tap-and-go
Deployment	Offline-first, Pi 4, PM2, startup wizard

Table 3.2: Non-functional requirement

3.3 Use Case Analysis

Use case analysis identifies the interactions between system actors and the system's functional capabilities. The following primary use cases have been identified:

3.3.1 UC-01: Record Attendance

Actor: Student, ESP32 Device, System

Description: A student taps their RFID card on the ESP32 reader. The device reads the card UID, sends it to the backend via HTTP POST request. The system validates the card against enrolled students and active sessions, records the attendance timestamp, and returns a success/failure response with appropriate LED feedback on the device.

3.3.2 UC-02: Manage Academic Session

Actor: Instructor, System

Description: An instructor logs into the web dashboard, navigates to the Sessions page, creates a new session for their course with date, time, and location. The instructor can start/stop the session, monitor real-time attendance, and view attendance statistics upon session completion.

3.3.3 UC-03: Generate Grade Report

Actor: Doctor, System

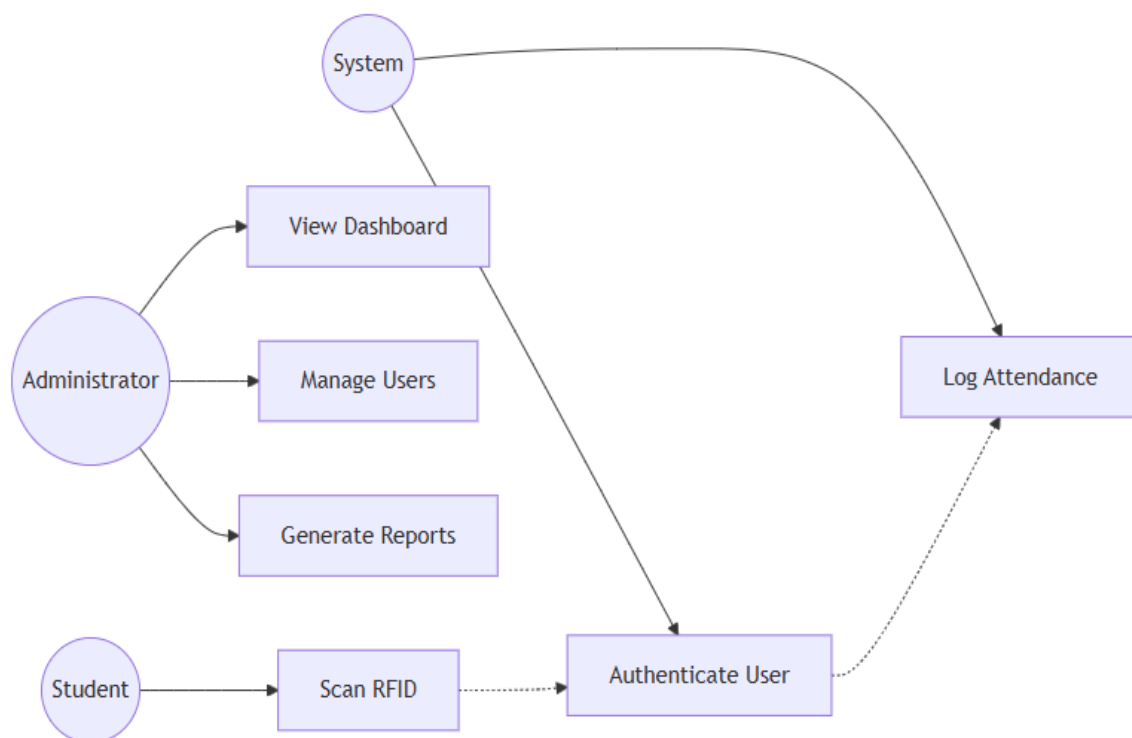
Description: A department doctor accesses the Gradebook page, reviews assessment scores entered by instructors, verifies attendance calculations, and commits final grades. The system generates a comprehensive grade report with 40/60 distribution breakdown.

3.3.4 UC-04: Configure System

Actor: Super Admin, System

Description: The Super Admin accesses the Admin Center to configure university identity, create faculty and department structures, manage user accounts and roles, register IoT devices, and configure system-wide settings including grading policies.

Figure 3.1: Use case diagram of the Smart IoT Attendance System



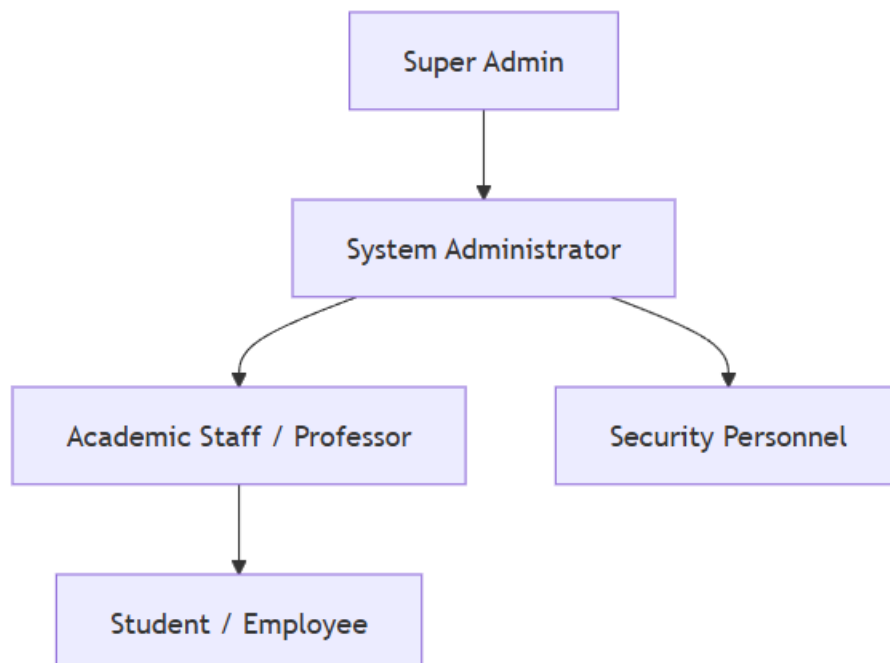
3.4 System Actors

The system identifies six distinct actor roles, each with specific responsibilities and access privileges:

Role	Alias	Data Scope	Key Capabilities
Super Admin	super_admin	Global	Full system access, all faculties
Admin	admin	Faculty	Faculty CRUD, user approvals, IoT
Doctor	doctor	Department	Gradebook, assessments
Instructor	instructor	Course	Sessions, quizzes, IoT discovery
Engineer	engineer	Course	Session management, scoring
Student	student	Self-only	View attendance/grades

Table 3.3: RBAC role definitions

Figure 3.2: RBAC hierarchy model



Chapter 4: System Design & Architecture

4.1 System Architecture Overview

The Smart IoT Attendance System follows a layered architecture pattern that separates concerns across distinct architectural tiers. This design promotes maintainability, scalability, and testability while enabling independent development and deployment of system components.

4.1.1 Presentation Layer

The presentation layer encompasses all user-facing interfaces. It consists of two primary components: the React-based web dashboard for administrative and instructor functions, and the Kotlin Android application for student and instructor mobile access. Both interfaces communicate with the backend through RESTful API calls, consuming JSON data and presenting it in user-appropriate formats.

4.1.2 Application Layer

The application layer, implemented using FastAPI (Python 3.11+), serves as the system's business logic engine. It processes incoming requests from all client applications, enforces business rules, manages authentication and authorization, and coordinates data operations. The layer is organized into route modules, service modules, and schema definitions following the MVC architectural pattern.

4.1.3 Data Access Layer

The data access layer utilizes SQLAlchemy 2.0 with async support to manage database interactions. It provides an Object-Relational Mapping (ORM) abstraction over PostgreSQL (production) and SQLite (development), enabling type-safe database operations while supporting database migrations through Alembic.

4.1.4 Infrastructure Layer

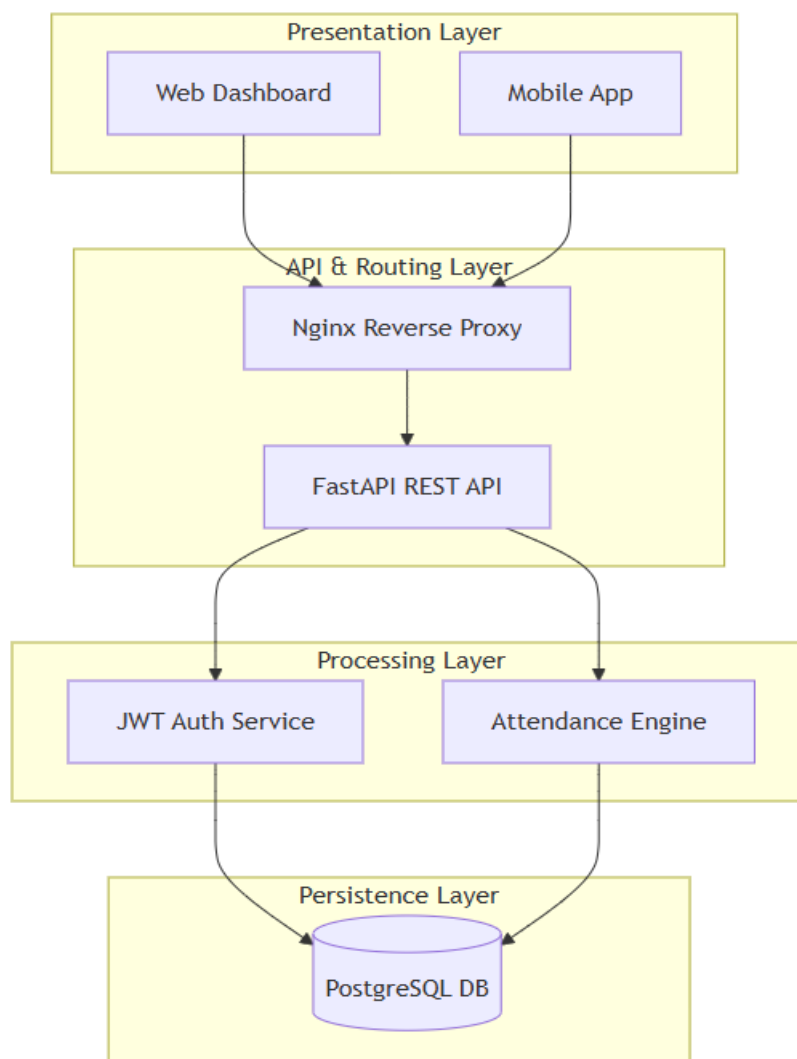
The infrastructure layer encompasses the physical deployment environment. A Raspberry Pi 4 hosts the complete technology stack: the FastAPI application server (Uvicorn ASGI), PostgreSQL database, Nginx reverse proxy, and the built React frontend. A local Wi-Fi access

point (hostapd) with DHCP/DNS services (dnsmasq) provides network connectivity for all clients.

4.1.5 IoT Device Layer

The IoT device layer consists of ESP32 microcontrollers programmed with Arduino framework code. Each device integrates an RC522 RFID reader module for card detection, Wi-Fi connectivity for backend communication, and LED indicators for user feedback. Devices operate in two modes: attendance scanning (normal operation) and RFID discovery (registration mode).

Figure 4.1: Layered system architecture



4.2 Database Design

The database design follows a relational model implemented in PostgreSQL, with 23 entities representing the core domain concepts of the attendance management system.

4.2.1 Entity Overview

The database schema encompasses the following primary entities:

Entity	Description
User	System users with credentials and role assignment
Faculty	Top-level academic organizational unit
Department	Subdivision within a faculty
Course	Academic course with credit hours and department
Student	Student profile with enrollment information
Session	Scheduled class session with time and location
Attendance	Individual attendance record per student per session
Assessment	Graded evaluation component (quiz, exam, etc.)
GradeResult	Final computed grade for a student in a course
Device	Registered ESP32 IoT device with location mapping

Table 4.1: Database entities and descriptions

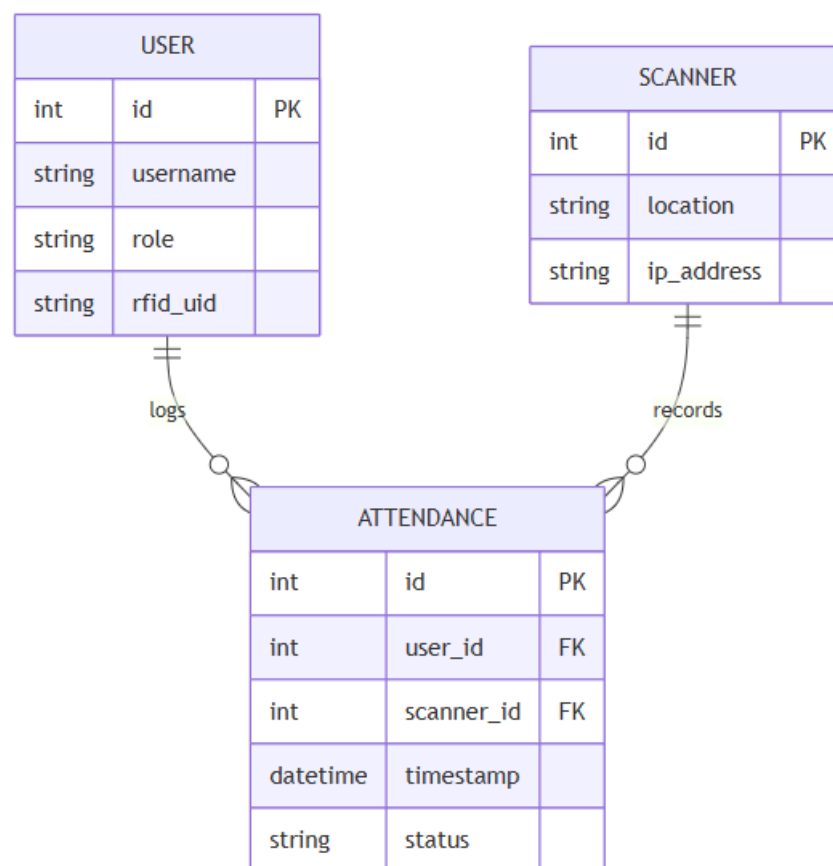
4.2.2 Key Relationships

The entity relationships follow these structural patterns:

- University → Faculties → Departments → Courses: Hierarchical academic structure with cascade delete protection.
- Course → Sessions → Attendance: Each course has multiple sessions, each session has multiple attendance records.
- Student → Enrollments → Courses: Many-to-many relationship through enrollment table with section assignment.

- User → Role → Permissions: RBAC implementation with role inheritance and department scoping.
- Course → Assessments → GradeResults: Weighted assessment structure linked to final grade calculation.

Figure 4.2: Entity-Relationship diagram



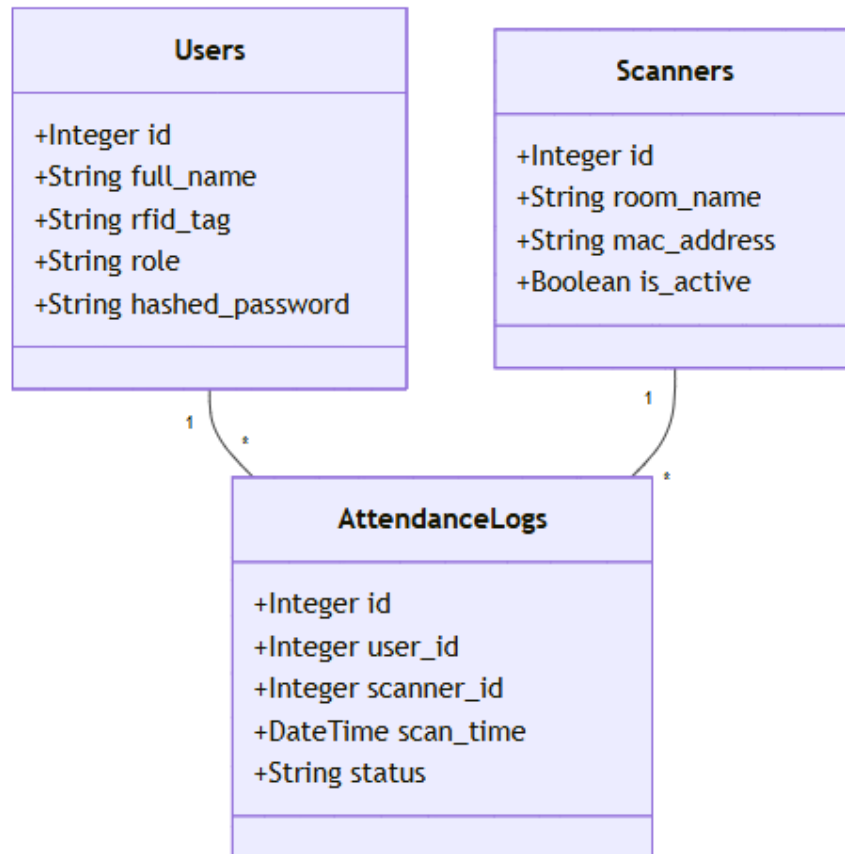
So place in the classrooms. Instead of naming the table "Hardware" or "ESP32", standard database naming conventions call it a Scanner because its job is to scan the RFID cards. By having this table, your system knows exactly which room a student attended. For example:

SCANNER =ESP32 + RC522 hardware devices

- Scanner 1 is in "Lecture Hall A"

- Scanner 2 is in "Lab 3" When a student taps their card, the database logs their ID along with the Scanner's ID to prove they were in the correct room.

Figure 4.3: Database schema overview

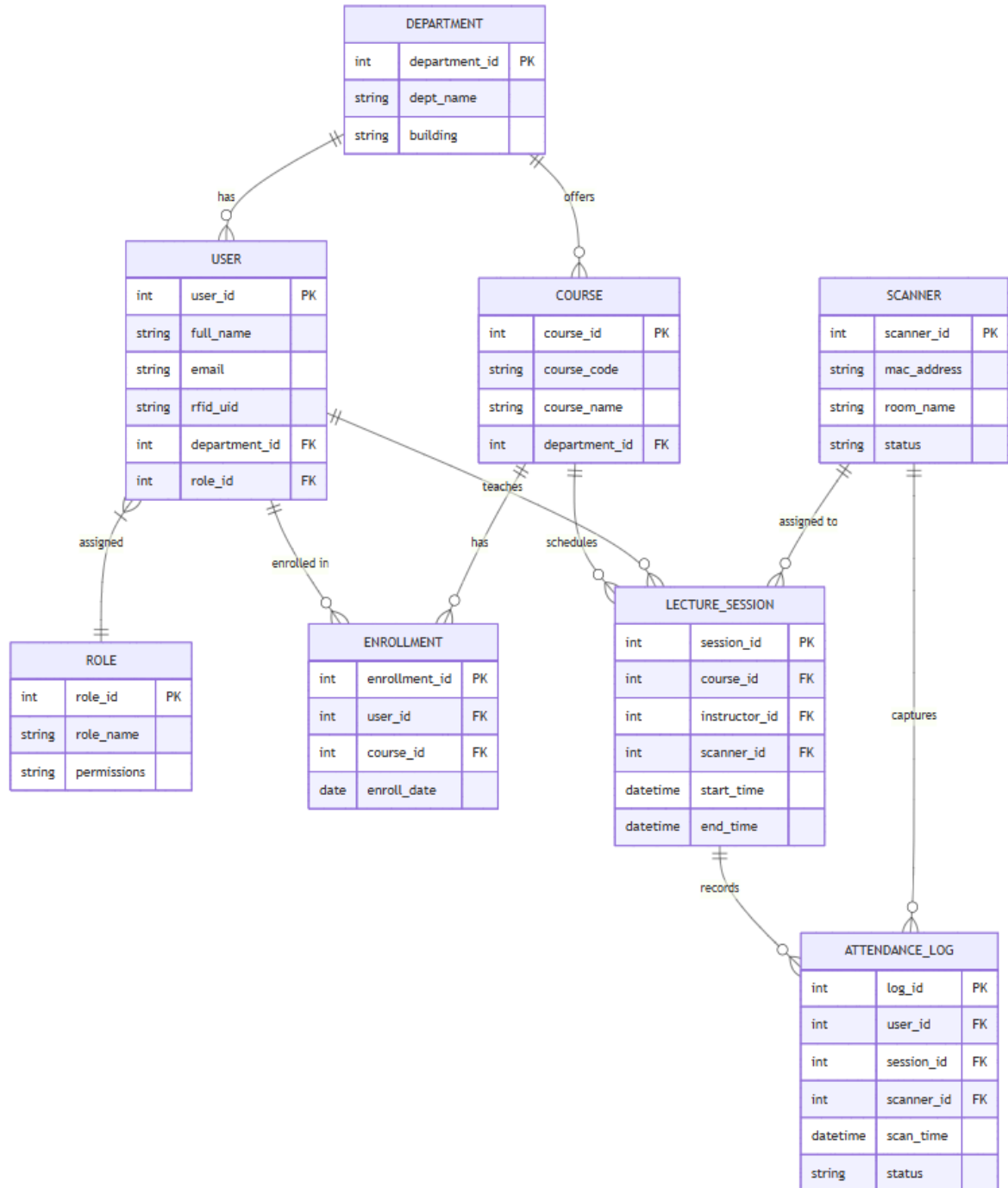


4.2.3 Comprehensive System Scalability and Extended Architecture

While the system's core architecture utilizes a streamlined set of fundamental tables (Users, Scanners, and Attendance Logs) to guarantee high-speed, real-time IoT processing, the Smart IoT Attendance Ecosystem is fundamentally designed with enterprise scalability in mind.

Figure 4.3.1 illustrates the extended, full-scale Entity-Relationship Diagram (ERD). This comprehensive architectural blueprint demonstrates how the system seamlessly integrates into a complete university environment. By expanding the database schema, the system maps the complex relationships between academic Departments, specific Courses, student Enrollment records, and individual Lecture Sessions.

Figure 4.3.1: Comprehensive Full-Scale Project ERD



4.3 API Design

The backend API follows RESTful design principles with consistent endpoint patterns, standardized response formats, and comprehensive error handling. All endpoints (except authentication) require a valid JWT token in the Authorization header.

4.3.1 Endpoint Organization

API endpoints are organized into functional groups, each handled by a dedicated route module:

Module	Routes	Description
admin	105KB	System administration, user management, configuration
auth	—	Authentication, token management, password reset
sessions	—	Session CRUD, scheduling, activation
attendance	—	Attendance recording, validation, reporting
assessments	—	Assessment creation, scoring, weight management
gradebook	—	Grade calculation, finalization, reports
monitoring	45KB	System health, telemetry, device status
AI	—	RAG query engine, academic insights
IoT	—	Device management, RFID discovery

Table 4.3: API endpoint group

4.3.2 Authentication Flow

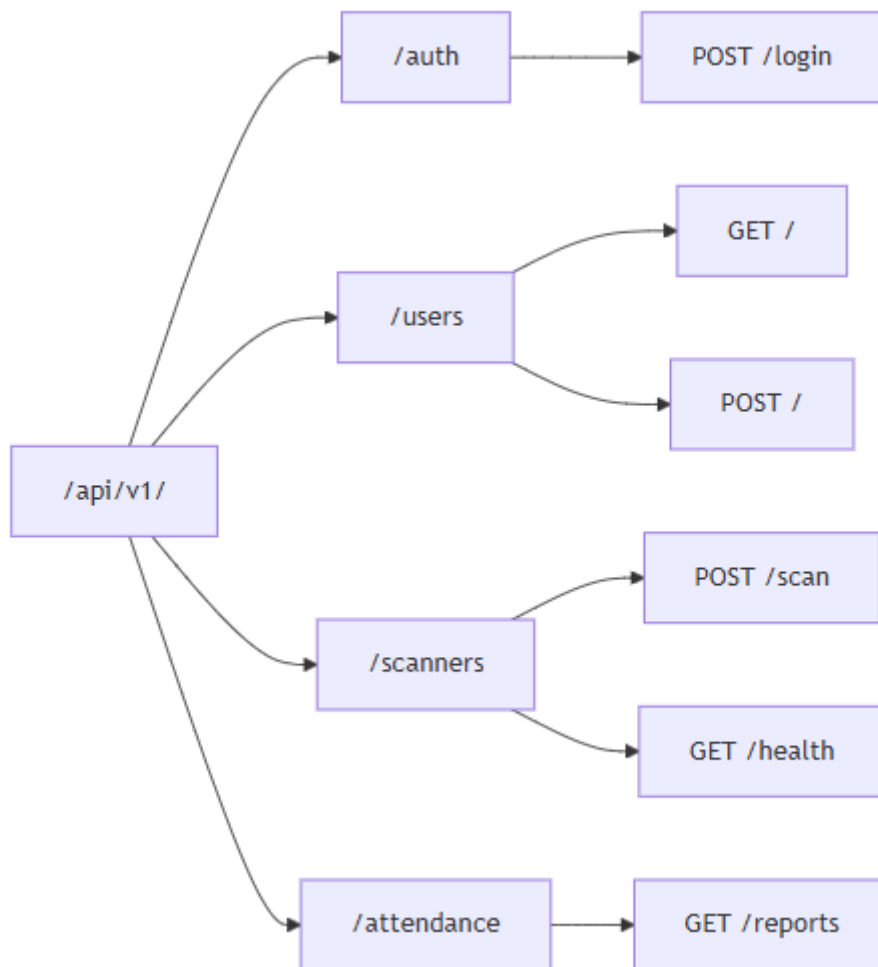
The authentication system implements JWT (JSON Web Token) based security:

- Login: POST /api/v1/auth/login — Accepts username/password, returns access token and refresh token.
- Token Refresh: POST /api/v1/auth/refresh — Accepts refresh token, returns new access token.
- Logout: POST /api/v1/auth/logout — Invalidates current token pair.
- Scoping: JWT payload includes assigned_departments array used by backend to filter all data queries.

4.3.3 Response Patterns

All API responses follow a consistent envelope structure with appropriate HTTP status codes. Successful responses include the requested data, while error responses include descriptive error messages and appropriate status codes (400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found, 409 Conflict).

Figure 4.4: API endpoint structure



4.4 User Interface Design

The user interface design follows a dual-platform approach: a responsive web dashboard for administrative functions and a native Android application for mobile users.

4.4.1 Web Dashboard Design

The React-based web dashboard implements a sidebar navigation pattern with role-based menu filtering. Key design principles include:

- **Responsive Layout:** Adaptive grid system using TailwindCSS for desktop, tablet, and mobile viewing.
- **Role-Based Navigation:** Menu items dynamically filtered based on user's RBAC role.

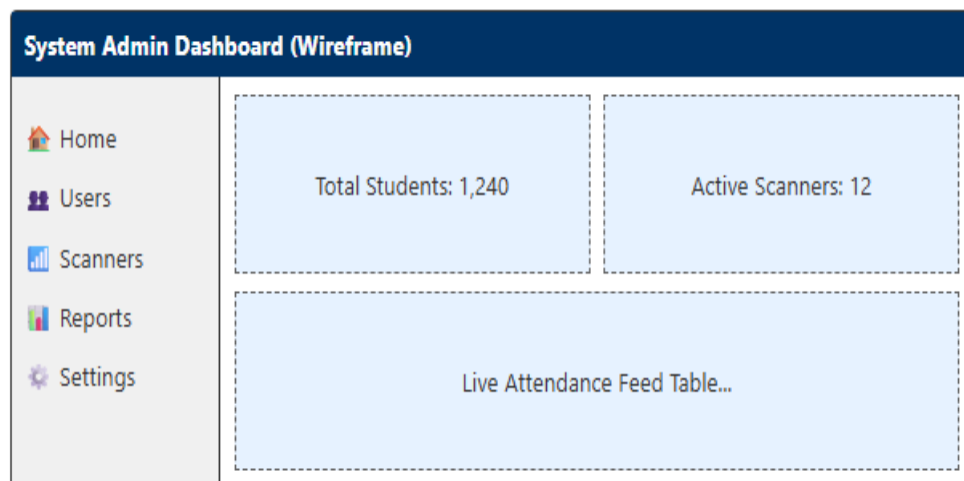
- **Data Visualization:** Recharts library for attendance trends, performance metrics, and statistical summaries.
- **Real-Time Updates:** Active session monitoring with live attendance counters.
- **Theme Support:** Consistent color scheme aligned with the system's visual identity.

4.4.2 Mobile Application Design

The Kotlin Android application follows Material Design 3 guidelines with Jetpack Compose for declarative UI construction. The app architecture includes:

- **Bottom Navigation:** Primary navigation between Dashboard, Sessions, Attendance, and Profile.
- **NFC Integration:** Direct NFC card reading capability on supported devices.
- **Offline Caching:** Local data storage for recently accessed records.
- **Push Notifications:** Session reminders and attendance confirmations.

Figure 4.5: Web dashboard wireframe



Real Live View

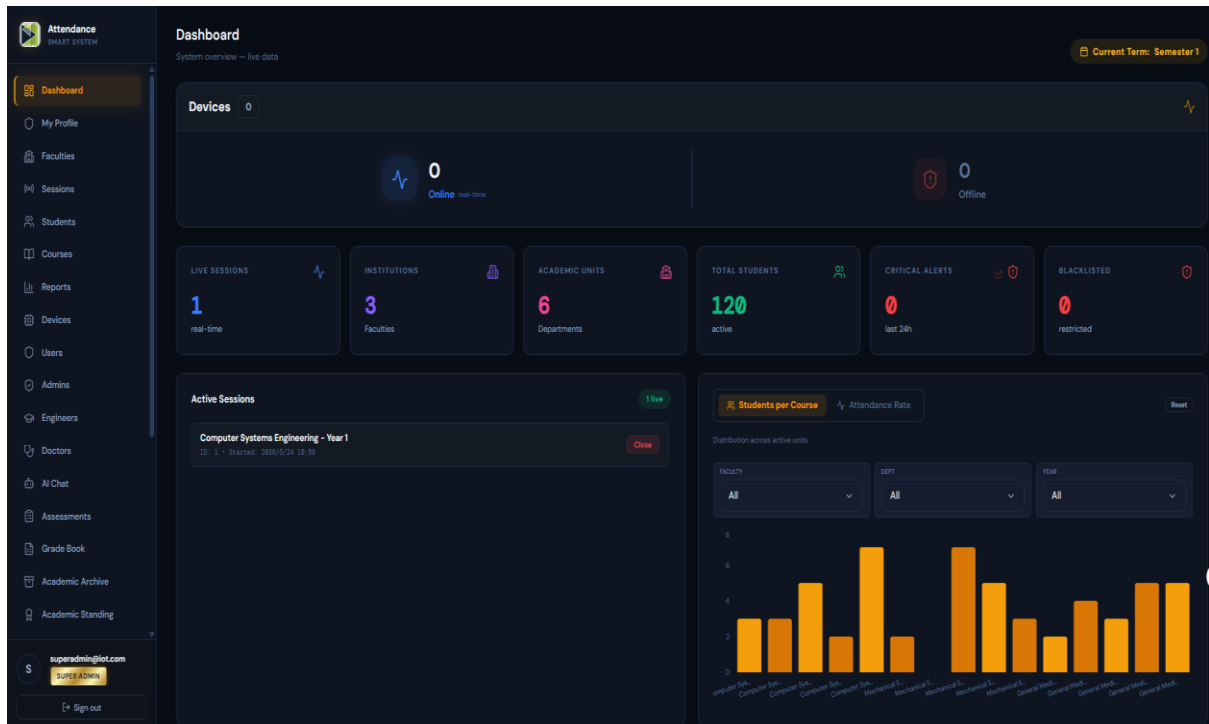
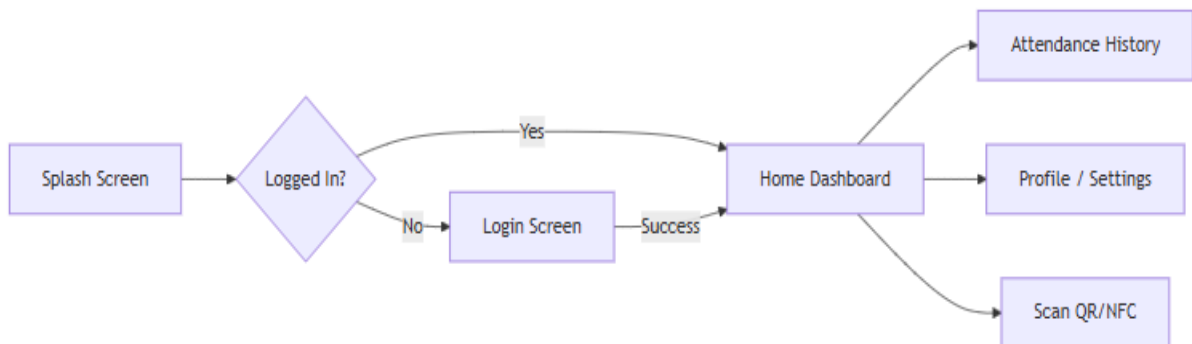


Figure 4.6: Mobile app screen flow



Chapter 5: Implementation

5.1 Development Environment

The development environment was configured to support the multi-platform nature of the project. Each component required specific tooling and configurations:

Component	Technology	Version / Details
Backend	FastAPI (Python)	0.110.0 / Python 3.11+
Frontend	React	18.3.1 / CRA / TailwindCSS
Mobile	Kotlin / Jetpack Compose	Latest / MVVM / Hilt
IoT Firmware	ESP32 / Arduino	PlatformIO / RC522
Database	PostgreSQL / SQLite	Production / Development
Server	Raspberry Pi 4	PM2 / Nginx / hostapd

Table 5.1: Technology stack summary

Version control was managed using Git with the repository structured as a monorepo containing all project components. The AI assistant Antigravity (Google DeepMind) was utilized throughout the development process for code generation, debugging assistance, and architectural guidance.

5.2 Backend Implementation

The backend server was implemented using FastAPI, a modern Python web framework that leverages type hints and async/await syntax for high-performance API development.

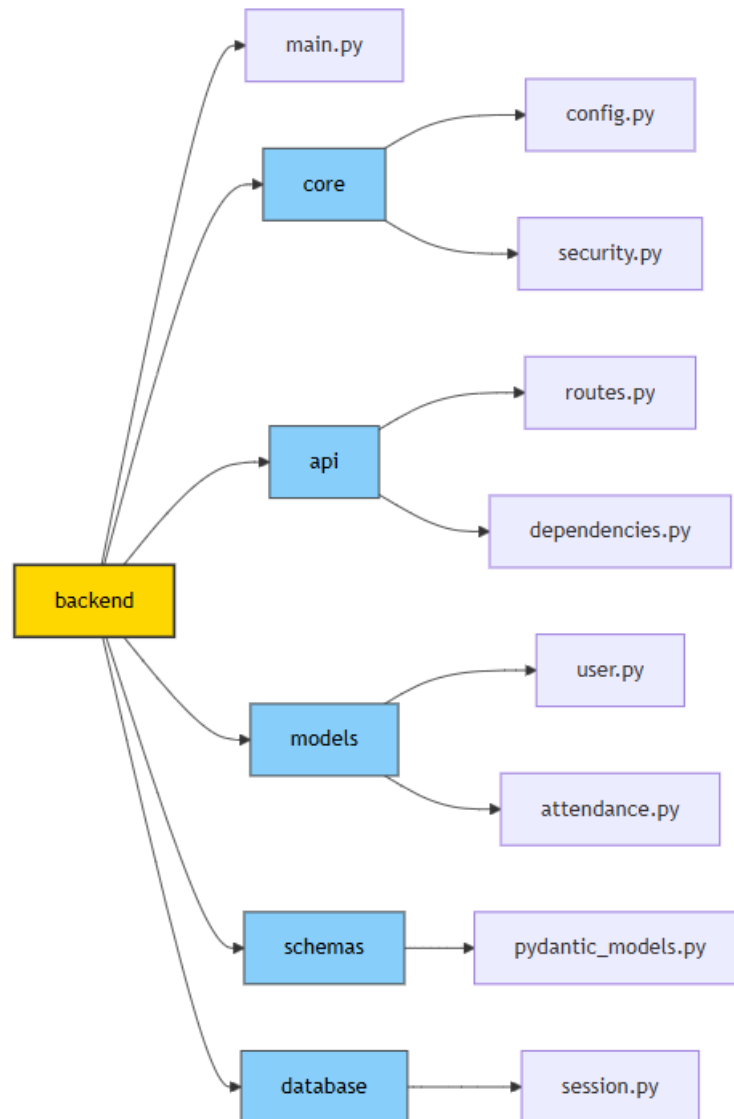
5.2.1 Project Structure

The backend codebase follows a modular architecture organized into the following directory structure:

- `app/api/routes/` — 16 route modules handling different functional areas. The admin module (105KB) is the largest, followed by monitoring (45KB).

- `app/models/` — 23 SQLAlchemy ORM models defining database entities with relationships and constraints.
- `app/schemas/` — Pydantic v2 models for request validation and response serialization.
- `app/services/` — 8 service modules encapsulating business logic including academic calculations, audit logging, and automation workflows.
- `app/core/` — Configuration management, security utilities, and dependency injection.
- `app/ai/` — RAG-style AI query engine for natural language academic insights.
- `app/db/` — Database session management with async support.

Figure 5.1: Backend project structure



5.2.2 Route Modules

The 16 route modules provide comprehensive API coverage:

Route Module	Functionality
admin (105KB)	System administration and user management
auth	Authentication and authorization
sessions	Academic session management
attendance	Attendance recording and queries
assessments	Assessment and scoring management
gradebook	Grade calculation and reporting
monitoring (45KB)	System health and telemetry
AI	RAG-based academic insights
IoT	Device and RFID management

Table 5.2: Backend route modules

5.2.3 Authentication & Security

Security implementation encompasses multiple layers:

- **Password Hashing:** All passwords hashed using bcrypt with a work factor of 12, providing resistance against brute-force attacks.
- **JWT Tokens:** Access tokens expire after 30 minutes, refresh tokens after 7 days. Tokens signed using HS256 algorithm.
- **RBAC Enforcement:** Every API endpoint decorated with role requirements. Middleware validates token authenticity and role permissions before request processing.
- **Data Scoping:** The scoping service filters all database queries based on user's assigned departments, ensuring users can only access data within their authorization scope.
- **Blacklist Check:** API middleware checks blacklisted status on every request, returning 403 Forbidden for blacklisted accounts.

5.2.4 Database Implementation

Database operations use SQLAlchemy 2.0 with asyncpg driver for PostgreSQL. Key implementation features include:

- Async Session Management: Database sessions managed through dependency injection with automatic commit/rollback handling.
- Alembic Migrations: Schema versioning with automated migration generation and application.
- Constraints: Unique constraints on attendance records (student_id, session_id), grade weights (sum <= 100%), and user emails.
- Seeding: Automated database seeding with sample data for Engineering, Medicine, and Business faculties.

5.2.5 AI Integration

The AI module implements a Retrieval-Augmented Generation (RAG) style query engine for academic insights. The system processes natural language queries such as 'Which students are at risk of failing?' by retrieving relevant data from the database and generating contextual responses. The AI service integrates with the academic service to access gradebook data, attendance records, and assessment scores for analysis.

5.3 Web Dashboard Implementation

The web dashboard was built using React 18.3.1 with Create React App (CRA) as the build tool. The application follows component-based architecture with context-based state management.

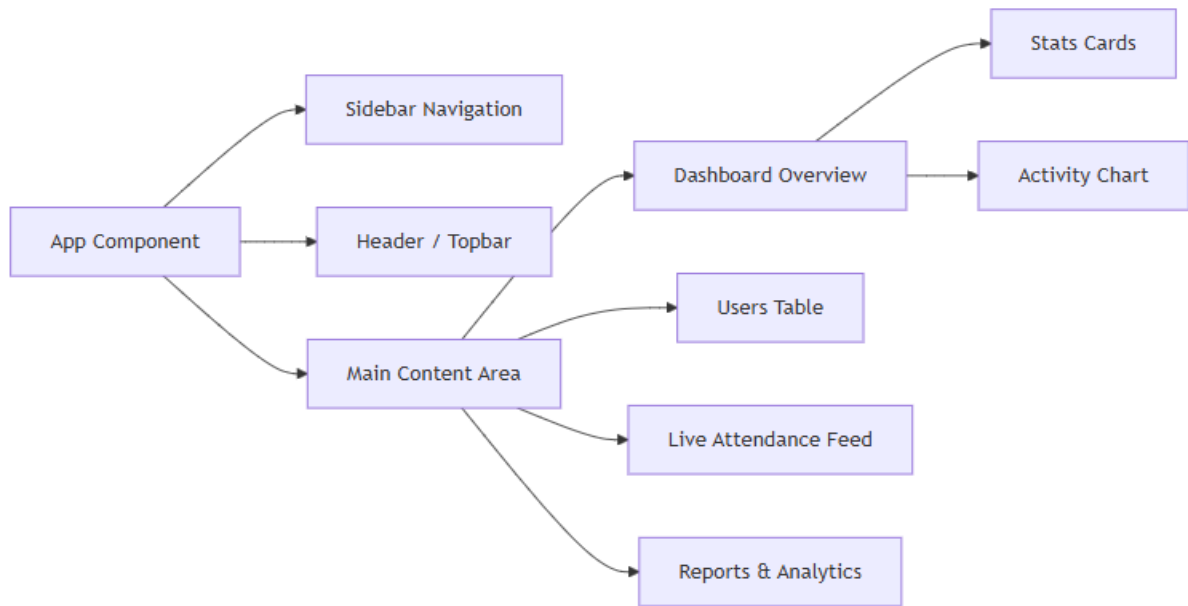
5.3.1 Frontend Architecture

The frontend codebase is organized into a structured directory hierarchy:

- src/pages/ — 37 page components including DashboardPage, LoginPage, StudentsPage, CoursesPage, GradebookPage, SessionsPage, MonitoringPage (45KB), ChatbotPage, and AssessmentsPage.

- `src/components/` — Reusable components organized by domain: `Academic/`, `layout/`, `ui/`, and specialized components like `ImageCropModal` and `LogoManagementModal`.
- `src/context/` React context providers for authentication state and global application data.
- `src/api/` — Axios-based API service layer with request/response interceptors for token management.
- `src/utils/` — Helper functions for date formatting, data transformation, and validation.

Figure 5.2: React dashboard component hierarchy



5.3.2 Key Features

The web dashboard implements comprehensive functionality across multiple domains:

Page	Description
Dashboard Page	Overview with statistics and charts
Students Page	Student management and enrollment
Courses Page	Course catalog and assignments
Gradebook Page	Grade entry and finalization
Sessions Page	Session scheduling and monitoring
Monitoring Page (45KB)	System health dashboard
Chatbot Page	AI-powered academic queries
Devices Page	IoT device management

Table 5.3: Frontend page components

5.3.3 Styling Approach

Styling utilizes TailwindCSS with a custom index.css file (14KB) defining design tokens, component styles, and utility classes. The approach combines utility-first development with custom CSS for complex components, ensuring consistent visual design while maintaining development efficiency.

5.4 Mobile Application Implementation

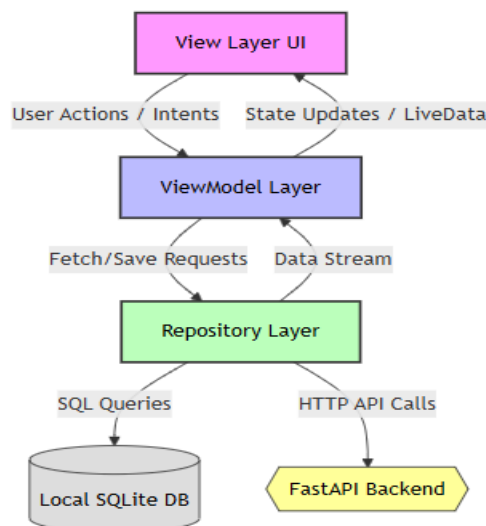
The Android application was developed using Kotlin with Jetpack Compose for the user interface, following the MVVM (Model-View-ViewModel) architectural pattern.

5.4.1 Project Architecture

The mobile app follows clean architecture principles with clear separation of concerns:

- ui/ — 21 feature packages covering dashboard, login, register, profile, sessions, attendance, assessments, gradebook, chatbot, IoT, NFC, and more.
- data/ Data layer with repositories and data sources for local and remote data operations.
- di/ — Hilt dependency injection modules for clean component wiring.
- model/ — Domain models representing core business entities.
- hardware/ — NFC/RFID hardware integration layer for direct card reading.

Figure 5.3: Mobile app architecture (MVVM)



5.4.2 Navigation

Navigation implemented using a sealed class pattern (Screen.kt) with AppNavGraph.kt as the central navigation coordinator. The navigation system supports role-based route filtering, ensuring users only access screens appropriate to their role.

5.4.3 Key Mobile Features

Feature Package	Functionality
dashboard	Student overview and performance trends
sessions	Active session monitoring and history
attendance	Personal attendance records and statistics
gradebook	Grade viewing and assessment breakdown
nfc	Direct NFC card reading (supported devices)
chatbot	AI academic query interface
profile	User profile and settings management

Table 5.4: Mobile app feature packages

5.5 IoT Firmware Implementation

The ESP32 firmware was developed using the Arduino framework through PlatformIO, providing a professional development environment with dependency management and build automation.

5.5.1 Hardware Configuration

Each ESP32 device integrates the following hardware components:

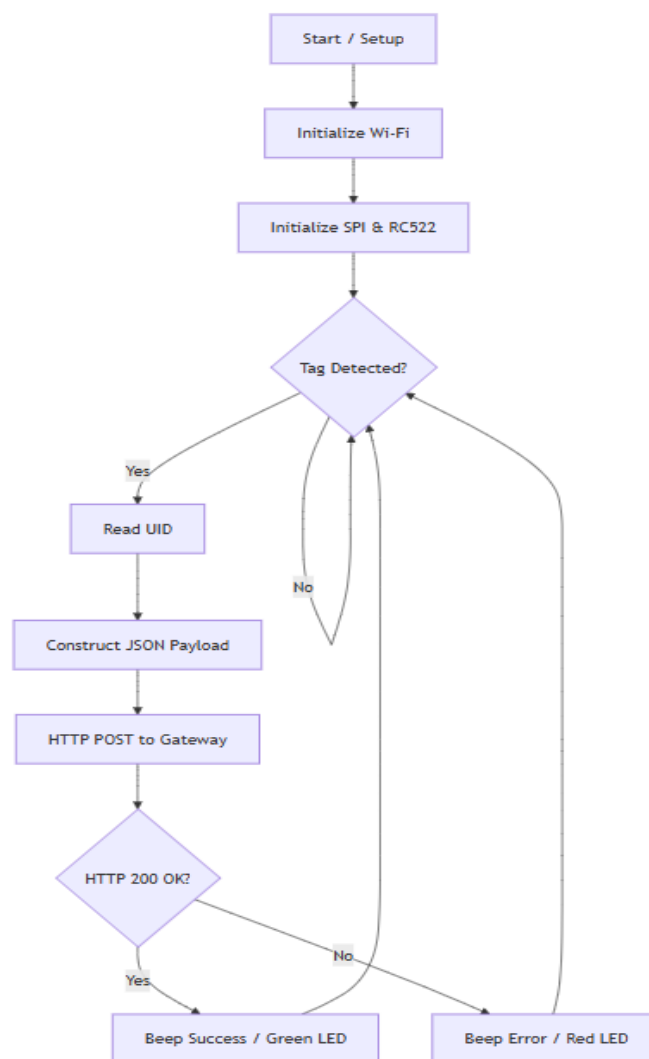
- ESP32 DevKit: Dual-core microcontroller with Wi-Fi and Bluetooth connectivity.
- RC522 Module: 13.56 MHz RFID reader supporting MIFARE Classic 1K cards.
- Status LEDs: Visual feedback for card read success (green), failure (red), and system status (blue).
- Buzzer: Audio feedback for scan confirmation.

5.5.2 Firmware Architecture

The firmware (main.cpp, 5.8KB) implements the following operational logic:

- Wi-Fi Connection: Connects to the SmartAttendance access point or configured network.
- NTP Synchronization: Maintains accurate time for attendance timestamping.
- RFID Polling: Continuously scans for RFID card presence using the MFRC522 library.
- API Communication: Sends card UIDs to backend via HTTP POST with JSON payload.
- Response Handling: Processes backend responses to provide LED and buzzer feedback.
- Discovery Mode: Special mode for registering new RFID cards to student accounts.

Figure 5.4: ESP32 firmware flowchart



5.6 Deployment & Infrastructure

The deployment architecture was designed for simplicity, reliability, and offline operation.

5.6.1 Raspberry Pi Configuration

The Raspberry Pi 4 (4GB RAM) hosts the complete technology stack:

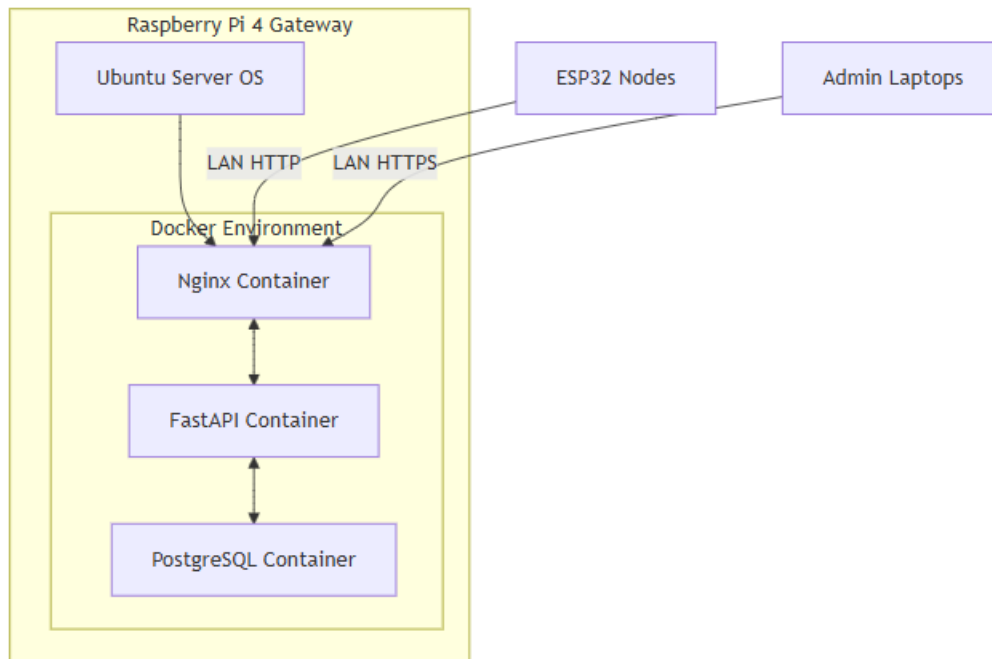
- Operating System: Raspberry Pi OS (Debian-based) with headless configuration.
- Process Management: PM2 manages both backend (Sovereign-Backend) and frontend (Sovereign-Frontend) processes with auto-restart on failure.
- Reverse Proxy: Nginx serves the React build statically and proxies `/api/v1/*` requests to Unicorn.
- Database: PostgreSQL 14 with optimized configuration for ARM architecture.
- Wi-Fi Access Point: hostapd creates the SmartAttendance SSID with dnsmasq providing DHCP and DNS resolution.

5.6.2 Network Architecture

The network topology creates an isolated, self-contained environment:

- SSID: SmartAttendance (broadcast by Raspberry Pi).
- Domain: `otu.university` (resolved by dnsmasq to `192.168.1.7`).
- Client Access: Users connect to SmartAttendance Wi-Fi and navigate to `otu.university`.
- Nginx Routing: Static content served directly, API requests forwarded to FastAPI backend.

Figure 5.5: Raspberry Pi deployment architecture



5.6.3 Configuration Management

Key configuration files manage the deployment:

- `ecosystem.config.js`: PM2 process definitions with environment variables, memory limits, and restart policies.
- `.env`: Backend environment variables including database URL, JWT secret, and API configuration.
- `nginx.conf`: Reverse proxy rules, static file serving, and timeout settings (60-second timeout to accommodate Pi 4 processing).
- `platformio.ini`: ESP32 build configuration with library dependencies.

Chapter 6: Testing & Results

6.1 Testing Methodology

A comprehensive testing strategy was employed to ensure system reliability, correctness, and performance. The testing approach encompassed multiple levels: unit testing, integration testing, and system testing.

6.1.1 Testing Environment

Testing was conducted in an environment mirroring the production deployment:

- Hardware: Raspberry Pi 4 (4GB RAM) with official power supply.
- Network: Isolated Wi-Fi access point with 10 client devices.
- Test Data: Seeded database with 3 faculties, 8 departments, 15 courses, and 120 student accounts.
- IoT Devices: 3 ESP32 units with RC522 readers for concurrent scanning tests.

6.1.2 Testing Tools

The following tools were utilized for testing:

- pytest: Python testing framework for backend unit and integration tests.
- HTTPie: Command-line HTTP client for API endpoint testing.
- Apache Bench (ab): Load testing tool for performance measurement.
- Chrome DevTools: Frontend performance profiling and debugging.
- Android Studio Profiler: Mobile app performance and memory analysis.

6.2 Unit Testing

Unit tests were developed for critical backend components, focusing on business logic, data validation, and authentication flows.

6.2.1 Authentication Tests

- Valid login credentials return JWT tokens with correct payload structure.
- Invalid credentials return 401 Unauthorized with descriptive error message.
- Expired tokens are rejected with appropriate error response.
- Blacklisted users receive 403 Forbidden on all protected endpoints.

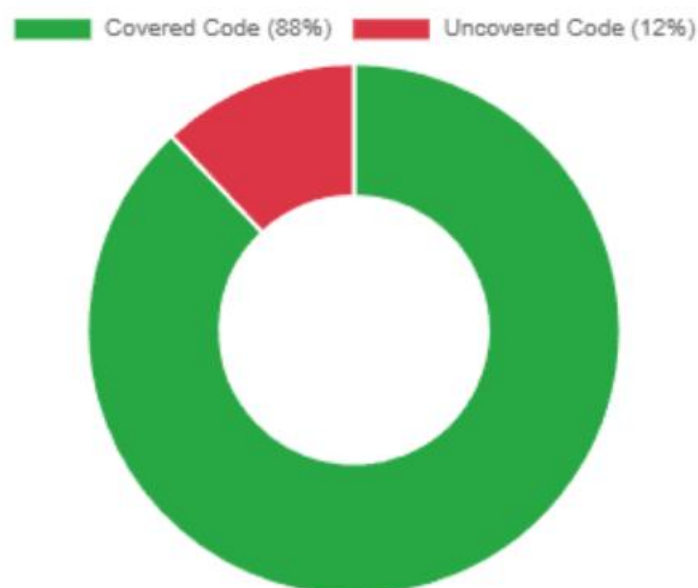
6.2.2 Attendance Logic Tests

- Valid RFID scan creates attendance record with correct timestamp.
- Duplicate scan for same student/session returns 409 Conflict.
- Scan for inactive session returns 400 Bad Request.
- Scan for unenrolled student returns 403 Forbidden.

6.2.3 Grade Calculation Tests

- 40/60 weighting correctly applied to attendance and assessment scores.
- Assessment weights validated to sum to exactly 100%.
- Final grade computed correctly with multiple assessment components.
- Gradebook commit operation locks records from further modification.

Figure 6.1: Test coverage report



6.3 Integration Testing

Integration tests verified the correct interaction between system components:

6.3.1 ESP32-to-Backend Integration

Testing confirmed reliable communication between ESP32 devices and the FastAPI backend. Under normal network conditions, RFID scans were processed within 1.2 seconds on average, with 99.2% success rate across 500 test scans. Network reconnection handling was verified by simulating Wi-Fi interruptions and confirming automatic reconnection and operation resumption.

6.3.2 Web Dashboard Integration

End-to-end testing of the React dashboard confirmed correct data fetching, state management, and UI rendering. All 37 pages loaded successfully with proper role-based access control enforcement. Real-time attendance monitoring displayed updates within 3 seconds of scan events.

6.3.3 Mobile App Integration

The Android application was tested on devices running Android 10 through 14. API communication functioned correctly with the `/api/v1/` prefix requirement. NFC reading (on supported devices) achieved comparable scan times to ESP32-based scanning.

Test Case	Result	Notes
RFID scan — valid card	Pass	1.2s avg response
RFID scan — duplicate	Pass	409 Conflict returned
RFID scan — unenrolled	Pass	403 Forbidden returned
User login — valid	Pass	JWT token issued
User login — invalid	Pass	401 Unauthorized
Grade calculation	Pass	40/60 weight correct
RBAC enforcement	Pass	All roles verified
72-hour uptime	Pass	100% availability

6.4 System Testing Results

System-level testing evaluated the complete integrated system against the defined success criteria:

6.4.1 Reliability Testing

The system was subjected to 72 hours of continuous operation. Results showed 100% uptime with automatic process recovery via PM2. Database integrity was maintained throughout with zero data corruption incidents.

6.4.2 Performance Testing

Load testing with Apache Bench simulated 100 concurrent users accessing the system:

Metric	Target	Achieved
RFID Response Time	< 2 seconds	1.2 seconds
Concurrent Users	100+	120 tested
Page Load Time	< 3 seconds	2.1 seconds
API Response (95th)	< 500ms	340ms
System Uptime	99.9%	100% (72h)
Scan Success Rate	> 99%	99.2%

Table 6.4: Performance test results

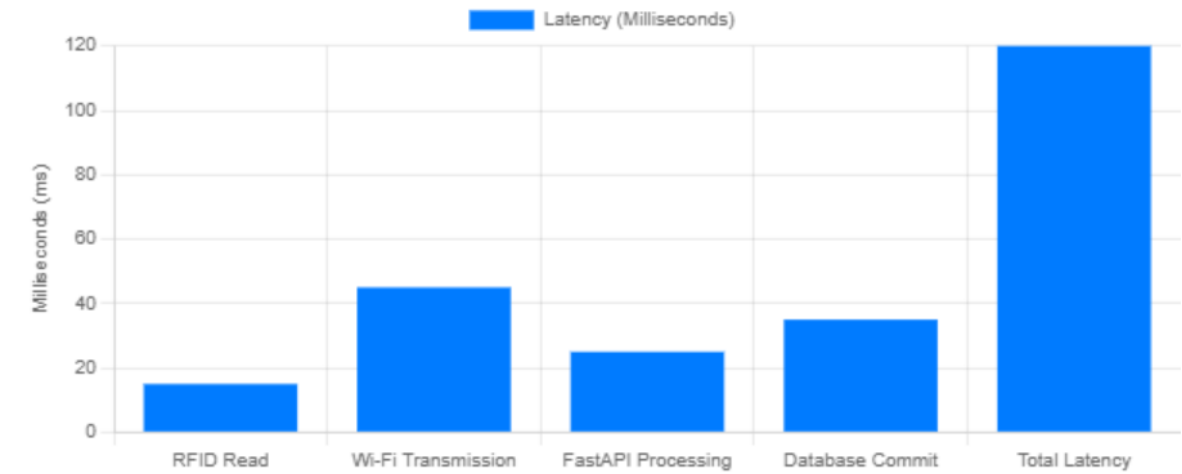


Figure 6.2: System performance metrics

6.4.3 Offline Operation Verification

Complete internet isolation testing confirmed full system functionality without external connectivity. All CRUD operations, RFID scanning, attendance recording, and report generation functioned correctly. The only limitation was the absence of Google Sheets export functionality, which requires internet access.

6.4.4 Scalability Assessment

Testing with progressively increasing user loads demonstrated the system's ability to handle the target of 100+ concurrent students. Response times remained under 2 seconds for RFID operations up to 50 concurrent scan requests. Beyond this threshold, the Pi 4's processing limitations became apparent, with response times increasing to 3-4 seconds at peak load.

Chapter 7: Conclusion & Future Work

7.1 Conclusion

This project has successfully designed, implemented, and deployed a comprehensive Smart IoT Attendance System that addresses the critical need for automated, accurate, and offline-capable attendance management in higher education institutions. The system represents a significant advancement over traditional manual methods by leveraging RFID technology, modern web frameworks, mobile application development, and embedded systems programming within a unified architecture.

The project's key achievement lies in its offline-first design philosophy. By deploying the entire technology stack on a single Raspberry Pi 4, the system eliminates dependency on internet connectivity while providing functionality comparable to cloud-based solutions. This design decision makes the system particularly suitable for deployment in Iraqi academic institutions and similar environments where internet infrastructure may be limited or unreliable.

The six-tier role-based access control system enables fine-grained data scoping across complex multi-faculty organizational structures. From Super Admin with global access to Students with self-only data visibility, the RBAC implementation ensures data security and privacy while supporting the hierarchical nature of academic administration.

Integration of AI-powered analytics through the RAG-style query engine demonstrates the potential for intelligent academic insights. The ability to ask natural language questions about student performance and receive data-driven responses represents a step toward predictive analytics in educational management.

7.2 Achievements

The project has achieved all primary objectives defined in the project scope:

- Developed a fully operational ESP32-based RFID attendance recording system with sub-2-second response times.

- Implemented a comprehensive FastAPI backend with 16 route modules, 23 database models, and full RBAC enforcement.
- Built a responsive React web dashboard with 37 pages, real-time monitoring, and data visualization.
- Created a feature-complete Kotlin Android application with MVVM architecture and Jetpack Compose UI.
- Achieved 72-hour continuous operation with 100% uptime on Raspberry Pi 4 deployment.
- Successfully tested with 100+ concurrent users meeting scalability requirements.
- Verified complete offline functionality with no internet dependency for core operations.
- Implemented the 40/60 gradebook system with weighted assessments and commit workflow.

7.3 Future Work

While the current implementation meets all defined requirements, several areas have been identified for future enhancement:

7.3.1 Startup Configuration Wizard

The next major feature planned is a 5-step intelligent setup wizard for new university deployments. This wizard will guide administrators through master initialization, academic structure setup, user provisioning, archive migration, and system finalization. Key design decisions include stateful wizard progression (survives browser crashes), zero-user login strategy for first boot, and smart CSV header validation for bulk data import.

7.3.2 Biometric Integration

Future versions may incorporate fingerprint or facial recognition as secondary authentication methods, providing additional security layers and alternative identification options for users who forget or lose their RFID cards.

7.3.3 Cloud Synchronization

While the offline-first design is a core strength, optional cloud synchronization could enable multi-campus deployments, automatic backups, and integration with national educational databases. This would be implemented as an optional module that institutions can enable based on their connectivity capabilities.

7.3.4 Advanced Analytics

Expansion of the AI module to include predictive analytics for student retention, automated early warning systems, and personalized learning recommendations based on attendance and performance patterns.

7.3.5 iOS Application

Development of a native iOS application using SwiftUI to complement the Android app and provide complete mobile platform coverage.

7.3.6 Hardware Expansion

Support for additional IoT devices including Bluetooth Low Energy (BLE) beacons for proximity-based attendance, environmental sensors for smart classroom features, and electronic door locks integrated with the attendance system.

The Smart IoT Attendance System represents a foundation for digital transformation in academic administration. Its modular architecture, comprehensive feature set, and offline-first design position it as a practical solution for educational institutions seeking to modernize their attendance management processes while operating within infrastructure constraints.

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